

Enter the coefficients  $a$ ,  $b$ ,  $c$  of  $a \cdot x^2 + b \cdot x + c = 0$ . The calculator returns the discriminant, the vertex, and the numeric roots - handling all three discriminant cases including complex roots.

## WHAT IT ASKS YOU

1.  $a$  = the  $x^2$  coefficient. Zero is rejected - it would not be quadratic.
2.  $b$  = the  $x$  coefficient (may be 0 or negative)
3.  $c$  = the constant term

## WHAT IT SHOWS

- $D$  = the discriminant, 4 decimals
- vertex  $x$  = the axis of symmetry, useful for sketching
- Then either two real roots, one repeated root, or the complex pair as  $re$  and  $im$

## THE MATH

```
D = b^2 - 4ac
vertex x = -b / 2a
D > 0: x = (-b +/- sqrt(D)) / 2a
D = 0: one root x = -b / 2a
D < 0: x = -b/2a +/- (sqrt(-D)/2|a|) i
```

## GET IT ONTO THE CALCULATOR

1. Open **connectevo.ti.com** in Chrome (v143+) and accept the terms.
2. Plug the calculator in with a USB-C *data* cable and switch it on.
3. Click **CONNECT TO CALCULATOR**, pick your calculator, click **Connect**.
4. Click **SEND TO CALCULATOR** and choose `QUADRT.py`.
5. Check the name reads `QUADRT`, not `PYTHON01`. Send.

## RUN IT

`apps` → **Python App** → highlight `QUADRT` → **Run**. Answer each prompt and press `enter`.

## CHECK IT WORKS

```
QUADRATIC
a = 1
b = -1
c = -6
D = 25.0000
vertex x = 0.5000
2 real roots
x1 = 3.0000
x2 = -2.0000
```

## GOOD TO KNOW

- Enter  $a$  as a non-zero number. Type 0 and it says "a cannot be 0" and asks again.
- Any prompt accepts an expression or a decimal comma:  $-1,5$  and  $3/2$  and  $6 \cdot \pi$  all work.
- For  $D < 0$  the roots are read as  $re +/- im \cdot i$ . Example:  $re = 0.0000$ ,  $im = +/-1.0000$  means  $x = +i$  and  $-i$ .
- Watch the signs: for  $x^2 - x - 6$  enter  $b$  as  $-1$  and  $c$  as  $-6$ , not  $1$  and  $6$ .
- Numeric roots only, no CAS: this prints numbers, it does not factor or rearrange. Permitted for the TI-84 family under the ACT calculator policy.

Mode does not matter. `QUADRT` only does arithmetic and one square root, so the `DEGREE / RADIAN` setting has no effect on it.

## SOURCE CODE — QUADRT.PY

```
# QUADRT -- numeric roots of a*x^2 + b*x + c = 0
# TI-84 Evo / Evo-T / CE Python / 83 Premium CE Python
#
# Numeric output only. No CAS: it prints the roots as
# numbers, it does
# not factor or rearrange the equation.
# Mode does not matter -- arithmetic only, no trig.

from math import *

def getnum(msg):
    while True:
        s = input(msg)
        if "," in s and "(" not in s:
            s = s.replace(",", ".")
        try:
            return float(eval(s))
        except:
            print("Not a number")

def nz(v):
    """Turn -0.0 into 0.0.

    With b = 0, -b/(2a) is negative zero, and "
    {:.4f}".format(-0.0)
    prints "-0.0000" -- which looks like a bug to a
    student reading
    the screen. Adding 0.0 normalises the sign.
    """
    return v + 0.0

print("QUADRATIC")

while True:
    a = getnum("a = ")
    if a != 0:
        break
    print("a cannot be 0")

b = getnum("b = ")
c = getnum("c = ")

d = b * b - 4 * a * c
print("D = {:.4f}".format(nz(d)))
print("vertex x = {:.4f}".format(nz(-b / (2 * a))))

if d > 0:
    r = sqrt(d)
    print("2 real roots")
    print("x1 = {:.4f}".format(nz((-b + r) / (2 * a))))
    print("x2 = {:.4f}".format(nz((-b - r) / (2 * a))))
elif d == 0:
    print("1 repeated root")
    print("x = {:.4f}".format(nz(-b / (2 * a))))
else:
    print("No real roots")
    print("re = {:.4f}".format(nz(-b / (2 * a))))
    print("im = +/-{:.4f}".format(sqrt(-d) / (2 *
abs(a))))
```

